STAT 516 - Homework 2 Answer Key

1. **(7 points)** Let E and F be events such that $P(E) = 0.60, P(F) = 0.50$, and $P(E^c \cap F^c) = 0.20$. Calculate $P(E \cup F^c)$.

$P(E \cup F) = 1 - P(E^c \cap F^c) = 1 - 0.20 = 0.80.$

Then $P(E \cap F) = 0.60 + 0.50 - 0.80 = 0.30$, so $P(E^c \cap F) = 0.50 - 0.30 = 0.20.$

Therefore, $P(E \cup F^c) = 1 - P(E^c \cap F) = 1 - 0.20 = 0.80.$

2. **(4 points)** You are given $P(A \cup B) = 0.75$ and $P(A \cup B^c) = 0.85$. Calculate $P(A)$.

$P(A \cup B) + P(A \cup B\hat{}c) = 1 + P(A)$. Therefore, $P(A) = 0.75 + 0.85 - 1 = 0.60.$

3. **(5 points)** A marketing survey indicates that 55% of the population owns an automobile, 35% owns a house, and 18% owns both an automobile and a house. Calculate the probability that a person chosen at random owns an automobile or a house, but not both.

P(auto or house, but not both)=0.55+0.35−2(0.18)=0.54.

4. **(5 points)** The probability that a visit to a primary care physician's office results in **neither** lab work nor referral to a specialist is 30%. Of those coming to the office, 35% are referred to specialists and 45% require lab work. Calculate the probability that a visit results in both lab work **and** referral to a specialist.

$P(L \cup R) = 1 - 0.30 = 0.70$. Therefore, $P(L \cap R) = 0.45 + 0.35 - 0.70 = 0.10.$

5. **(5 points)** Among a large group of patients recovering from knee injuries, 18% visit **both** a physical therapist and a chiropractor, whereas 15% visit **neither**. The probability that a patient visits a chiropractor **exceeds** the probability that a patient visits a physical therapist by 0.10. Calculate the probability that a randomly chosen patient visits a physical therapist.

Let $x = P$(physical therapist). Then P(chiropractor)=$x + 0.10$. Since $P(P \cup C) = 1 - 0.15 = 0.85$, $x + (x + 0.10) - 0.18 = 0.85$. Thus, $2x - 0.08 = 0.85$, so $x = 0.465$.

6. (7 points) Under an insurance policy, a maximum of five claims may be filed per year. Let $P(n)$ be the probability that a policyholder files n claims during a year, where n=0,1,2,3,4,5. Suppose:

(a) $P(n) \geq P(n+1)$ for $n = 0,1,2,3,4$.

(b) The difference between $P(n)$ and $P(n+1)$ is the same for all n.

(c) Exactly 50% of policyholders file fewer than two claims.

Calculate the probability that a randomly selected policyholder files more than three claims.

Let $d = P(1) - P(0)$, $P(n) = a - nd$ for $n = 0,1,2,3,4,5$. Since $P(0) + P(1) = 0.50$, $2a - d = 0.50$. Also, the probabilities sum to 1, so $6a - 15d = 1$. Solving gives $d = 1/24$ and $a = 13/48$. More than three claims means 4 or 5 claims, so $P(4) + P(5) = (a - 4d) + (a - 5d) = 1/6$.

7. (7 points) An insurance agent offers auto insurance, homeowners insurance, and renters insurance. Homeowners insurance and renters insurance are mutually exclusive. The profile of the agent's clients is as follows:

(a) 25% have none of these three products.

(b) 55% have auto insurance.

(c) Twice as many clients have homeowners insurance as renters insurance.

(d) 25% have exactly two of these products.

(e) 10% have homeowners insurance but not auto insurance.

Calculate the percentage of clients who have both auto and renters insurance.

Let P(renters)=x, so P(homeowners)=2x. Since 25% have none, 75% have at least one. Also, exactly two products means $P(A \cap H) + P(A \cap R) = 0.25$. Thus $0.75 = 0.55 + 3x - 0.25$, so $x = 0.15$ and $P(H) = 0.30$. Since 10% have homeowners but not auto, $P(A \cap H) = 0.30 - 0.10 = 0.20$. Therefore, $P(A \cap R) = 0.25 - 0.20 = 0.05$.

8. (5 points) The probability that a homeowner files a liability claim is 0.05, and the probability that the homeowner files a property claim is 0.12. The probability that the homeowner files a liability claim but not a property claim is 0.02. Calculate the probability that the homeowner files neither type of claim.

$P(L \cap P) = P(L) - P(L \cap P^c) = 0.05 - 0.02 = 0.03$.

Then $P(L \cup P) = 0.05 + 0.12 - 0.03 = 0.14$. Therefore, $P(neither) = 1 - 0.14 = 0.86$.

9. (7 points) An urn contains 12 balls: 5 red and 7 blue. A second urn contains 15 red balls and an unknown number of blue balls. One ball is drawn from each urn. The probability that both balls are the same color is 0.50. Calculate the number of blue balls in the second urn.

Let x be the number of blue balls in the second urn. Then $(5/12)(15/(15+x)) + (7/12)(x/(15+x)) = 0.50$. Thus $(75+7x)/(12(15+x)) = 0.50$, so $75 + 7x = 90 + 6x$, and $x = 15$.

10. (6 points) The probability of event U is 0.75 and the probability of event V is 0.45. What is the lowest possible value of $P(U \cap V)$?

$P(U \cap V)$= P(U)+P(V)- P(U∪ V)

The lowest possible value is max(0, P(U)+P(V)−1)=max(0, 0.75+0.45−1)=0.20.

11. (7 points) Roll a fair die. Let A be the event that the outcome is even. Let B be the event that the outcome is 4 or smaller. Let C be the event that the outcome is 3 or larger. Are A and B independent? Are B and C independent?

A={2,4,6}, B={1,2,3,4}, and C={3,4,5,6}. For A and B, P(A ∩ B)=2/6=1/3 and P(A)P(B)=(1/2)(4/6)=1/3, so A and B are independent. For B and C, P(B ∩ C)=2/6=1/3, but P(B)P(C)=(4/6)(4/6)=4/9. Since 1/3 ≠ 4/9, B and C are not independent.

12. (7 points) A matching pair of blue gloves, a matching pair of red gloves, and a matching pair of green gloves are in a drawer. A person blindly pulls two gloves out of the drawer simultaneously. Let B be the event that at least one glove is blue and A be the event that both gloves are blue. Given that B occurs, find the probability of A.

Answer 1: There are $\binom{6}{2}$ =15 total pairs of gloves, P(B)=[$\binom{2}{1}\binom{4}{1} + \binom{2}{2}]/\binom{6}{2}$ =9/15. P(A)= $\frac{\binom{2}{2}}{\binom{6}{2}} = \frac{1}{15}$,

P(A|B)=$\frac{P(A \cap B)}{P(B)} = \frac{1}{9}$.

Answer 2: There are $\binom{6}{2}$ =15 total pairs of gloves. The number of pairs with no blue gloves is $\binom{4}{2}$=6, so the number of pairs with at least one blue glove is 15−6=9. Only one pair contains both blue gloves. Therefore, P(A|B)=1/9.

13. (5 points) Roll a pair of dice. Let B be the event that the two dice have different values and A be the event that the sum of the dice is even. Given that B occurs, find the probability of A.

There are 30 ordered outcomes with different dice values. Among these, the sum is even when both dice are odd or both dice are even, but different. There are 3·2+3·2=12 such outcomes. Therefore, $P(A|B) = 12/30 = 2/5$.

14. (6 points) At a certain college, 35% of students live in a residence hall, and 65% live off campus. Students who live in a residence hall arrive to class on time with probability 0.90, while students who live off campus arrive on time with probability 0.75. If a randomly selected student arrives on time, what is the probability that the student lives in a residence hall?

By Bayes' theorem, $P(R|T) = 0.35(0.90)/[0.35(0.90) + 0.65(0.75)] = 0.315/0.8025 = 42/107.$

15. (5 points) Let A, B, C be three events. You are given $P(A) = 0.40, P(B) = 0.25, P(C|A) = 0.50$, and $P(C|B) = 0.60$. You are also given that $C \cap A$ and $C \cap B$ are mutually exclusive. Calculate $P(C \cap (A \cup B))$.

$P(C \cap A) = P(C|A)P(A) = 0.50(0.40) = 0.20.$ $P(C \cap B) = P(C|B)P(B) = 0.60(0.25) = 0.15.$
Since $C \cap A$ and $C \cap B$ are mutually exclusive, $P(C \cap (A \cup B)) = 0.20 + 0.15 = 0.35.$

16. (6 points) Let A, B, C be three events. The events are pairwise independent but not mutually independent. The probability of each event is 0.40, and the probability that all three occur is 0.08. Calculate the probability that none of them occur.

Since the events are pairwise independent, $P(A \cap B) = P(A \cap C) = P(B \cap C) = 0.40(0.40) = 0.16$. By inclusion-exclusion, $P(A \cup B \cup C) = 3(0.40) - 3(0.16) + 0.08 = 0.80$. Therefore, P(none)=1−0.80=0.20.

17. (6 points) A box contains 5 red balls and 7 white balls. A sample of size 3 is drawn without replacement. What is the probability of obtaining 1 red ball and 2 white balls, given that at least 2 of the balls in the sample are white?

P(1R,2W | at least 2W) = $\left[\binom{5}{1}\binom{7}{2}\right] / \left[\binom{5}{1}\binom{7}{2} + \binom{7}{3}\right] = 105/(105 + 35) = 3/4.$